Geochimica et Cosmochimica Acta Vol. 50, pp. 2799–2813
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0016-7037/86/$3.00 + .00

Compositional differences in enstatite chondrites based on carbon and nitrogen stable isotope measurements

MONICA M. GRADY, I. P. WRIGHT, L. P. CARR and C. T. PILLINGER
Department of Earth Sciences, The Open University, Walton Hall, Milton Keynes MK7 6AA, England

(*Received February* 4, 1986; *accepted in revised form September* 18, 1986)

Abstract—Carbon and nitrogen abundance and isotopic compositions, from four EH4, one EH5, five EL6 chondrites and one aubrite, were determined by using stepped pyrolysis (N only) and combustion (N and C) extractions in attempts to distinguish the components present. Carbon contents range from 0.15 to 0.70 wt%, with no systematic relationship between carbon content and meteorite group or petrologic type. Whole-rock $\delta^{13}C$ values range from −28.5 to −4.1‰. Most C occurs as graphite, and when temperature steps above 700°C are considered, there is a difference between EH4,5 ($\delta^{13}C$ = −9.1 to −5.8‰) and EL6 chondrites ($\delta^{13}C$ = −6.7 to +4.2‰). Carbon in Bustee aubrite is isotopically lighter ($\delta^{13}C$ = −24‰) than in any enstatite chondrite.

Nitrogen occurs as osbornite, sinoite and in isostructural substitution for oxygen in silicate lattice sites. Nitrogen abundances and isotopic compositions are more variable than C, due to the heterogenous distribution of N-bearing minerals. Three EL6's containing osbornite have higher N concentrations than other type 6 enstatite chondrites. Sinoite, where present, is depleted in ^{15}N relative to osbornite. Nitrogen in the Bustee aubrite has a similar abundance and $\delta^{15}N$ value to those of EL6's, again dominated by the presence of osbornite.

In addition to the refractory C- and N-bearing minerals there is also organic material (largely terrestrial contamination) and evidence for at least two "exotic" components. The first is a host for Xe (HL), and is characterized by $\delta^{13}C \leq -47$‰ and $\delta^{15}N \leq -73$‰, whereas the second is less well-defined, but is marked by $\delta^{15}N$ = +269‰.

INTRODUCTION

ENSTATITE CHONDRITES compose the smallest group of chondritic meteorites, numbering only twenty five in all, of which thirteen are observed falls (GRAHAM *et al.*, 1985). They exhibit textural variations similar to carbonaceous and ordinary chondrites, although differing significantly in mineralogy and mineral chemistry, being of a more reduced nature. Low calcium pyroxene dominates the composition of enstatite chondrites; the only other major mineral components present are iron-nickel metal (in some cases silica-rich) and nickel-poor troilite. Several unusual minor constituents have also been identified, *e.g.* oldhamite (CaS), daubreelite ($FeCr_2S_4$), sinoite (Si_2N_2O) (see MASON, 1966, and KEIL, 1968, for more complete mineralogical descriptions). Enstatite chondrites exist as petrologic types 3 to 6 (VAN SCHMUS and WOOD, 1967), and have been divided into two groups (BAEDECKER and WASSON, 1975). SEARS *et al.* (1982) named the two sets "EH" and "EL", by analogy with the high- and low-iron groups of the ordinary chondrites. Petrologic types 3, 4 and 5 fall into the EH group, (the sole exception being RKPA 80259, which is an EL5 (SEARS *et al.*, 1984)), whereas all type 6's are EL.

The location within the solar nebula where E-chondrites formed remains controversial, as does the formation mechanism. The highly reduced state of E-chondrites suggest that they probably originated in a region of the solar nebula with either (i) a high C/O ratio (>0.85, as opposed to the more usual value of 0.6, LARIMER, 1968, 1975; LARIMER and BARTHOLOMAY, 1979); (ii) a high H/O ratio (at least five times greater than the solar value, BAEDECKER and WASSON, 1975); or (iii) high pressures (HERNDON and SUESS, 1976; SEARS, 1980). The oxygen isotopic composition ($\delta^{17}O$, $\delta^{18}O$) of E-chondrites shows them to be unique amongst meteorite groups, with values close to those of the bulk Earth-Moon system (CLAYTON and MAYEDA, 1978; CLAYTON *et al.*, 1984).

In terms of formation mechanism, two competing hypotheses have been established: (i) differential accretion temperatures, leading to incomplete condensation of volatiles (*e.g.* LARIMER, 1973; ALAERTS and ANDERS, 1979), and (ii) volatile loss due to progressive thermal metamorphism (BINZ *et al.*, 1974; IKRAMUDDIN *et al.*, 1976; BISWAS *et al.*, 1980). The most recent trace element studies have failed to resolve this issue (HERTOGEN *et al.*, 1983).

Appraisal of the mineralogy of E-chondrites suggests that all samples (except some recently discovered unequilibrated enstatite chondrites, PRINZ *et al.*, 1984; WEEKS and SEARS, 1985) have been heated to temperatures above 600°C, whilst the presence of tridymite in EL6's shows that E-chondrites of this type have been heated to over 870°C (DODD, 1981). Despite these apparently high temperatures of condensation/metamorphism, all enstatite chondrites are rich in volatiles, EH4 chondrites possessing volatile trace element (*i.e.* Se, Ag, Zn, In, Cd, Br, etc.) contents close to those of CI1 chondrites (HERTOGEN *et al.*, 1983). Furthermore, "mysterite", a volatile-rich carbonaceous component thought to be a late-stage primary nebula condensate (LAUL *et al.*, 1973), has been found in the Abee enstatite chondrite (GANAPATHY and LARIMER, 1980). The primitive nature of E-chondrites is also supported by

the presence in types 3 to 5 of primordial noble gases, as well as more exotic components such as Xe (HL) (CRABB and ANDERS, 1981, 1982).

The abundance of carbon in E-chondrites is high, comparable to concentrations in C3 meteorites. Carbon contents for EH4 chondrites range from 0.22 to 0.56 wt%. EH5 chondrites all fall within this range, with abundances from 0.3 to 0.54 wt%, while EL6 meteorites are more variable, containing from 0.06 to 0.84 wt% C (BOATO, 1954; MOORE and LEWIS, 1965, 1966; WIIK, 1969; BELSKY and KAPLAN, 1970; OTTING and ZAHRINGER, 1970; YANG and EPSTEIN, 1983). The distribution of carbon in enstatite meteorites is somewhat different from that in either carbonaceous or ordinary chondrites. Graphite is ubiquitous, occurring in two different associations (i) with silicate minerals and (ii) in metal grains. KEIL (1968) reported that more graphite occurred in Type II chondrites (*i.e.* EL6) than Type I (*i.e.* EH4,5). Cohenite (Fe_3C) has been identified in E-chondrites (MASON, 1966; KEIL, 1968), but, as detailed by DEINES and WICKMAN (1985), the evidence is very scant; the most reliable report is for Abee (HERNDON and RUDEE, 1978). Small amounts of organic compounds have also been found in Abee (BELSKY and KAPLAN, 1970).

Despite the high carbon abundances in E-chondrites, few determinations of carbon isotopic composition have been made. BOATO (1954) reported the first $\delta^{13}C$ value for any E-chondrite (−12.9‰ in Indarch). Subsequent measurements on EH4's have yielded a range of $\delta^{13}C$ values from −12.9 to −3.2‰. The $\delta^{13}C$ value of carbon from the EH5 chondrite St. Mark's (−6.2‰) falls within the range of compositions from EH4 meteorites, whereas EL6 chondrites show a wider range in isotopic compositions ($\delta^{13}C$ = −7.3 to +9.0‰), but generally enriched in ^{13}C relative to the EH4,5 group (BELSKY and KAPLAN, 1970; CARR *et al.*, 1983; YANG and EPSTEIN, 1983; DEINES and WICKMAN, 1985).

Nitrogen, like carbon, is present in E-chondrites as reduced species, such as nitrides. Two have been petrographically identified: osbornite (TiN, first identified in the Bustee aubrite: STORY-MASKELYNE, 1870) and sinoite (silicon oxynitride, Si_2N_2O), a mineral formed at high temperatures and low oxygen pressures, (RYALL and MUAN, 1969). Another mode of occurrence of nitrogen could be the substitution of N^{3-} for O^{2-} in the silicate lattice (BAUR, 1972). E-chondrites possess variable nitrogen abundances—EH4: 100–540 ppm; EH5: 180–275 ppm; EL6: 40–850 ppm (MOORE and GIBSON, 1969, MOORE *et al.*, 1969; INJERD and KAPLAN, 1974; KUNG and CLAYTON, 1978). Much of the variation may be due to sampling problems, dependent on the distribution of nitrides. Such minerals are more abundant in some EL6's, accounting for the wider range in nitrogen concentrations in this type. Indeed, a density separate (theoretically sinoite-rich) from the Hvittis EL6 chondrite had a nitrogen concentration of 2674 ppm (KUNG and CLAYTON, 1978). Considerable heterogeneity in nitrogen distribution within a single meteorite has been demonstrated by THIEMENS and CLAYTON (1983) in their study of three individual clasts from the Abee EH4 brecciated chondrite, where nitrogen contents varied from 254 to 853 ppm.

Previous reports of nitrogen isotopic data for E-chondrites are very sparse, (INJERD and KAPLAN, 1974; KUNG and CLAYTON, 1978; THIEMENS and CLAYTON, 1983; CARR *et al.*, 1983; CARR and PILLINGER, 1984). The $\delta^{15}N$ values of EH4 chondrites lie in a narrow range between −31 and −26‰, the only exceptions being samples from which nitrogen was not extracted quantitatively (KUNG and CLAYTON, 1978). Nitrogen isotopic compositions of EL6 chondrites are more varied, from −44 to −24‰, possibly due to the heterogeneous distribution of sinoite within these meteorites. It is notable that Hvittis, a specimen in which sinoite has been identified, has the lightest $\delta^{15}N$ value: the sinoite-rich separate analyzed by KUNG and CLAYTON (1978) had a $\delta^{15}N$ value of −44‰, implying that sinoite is enriched in ^{14}N relative to other nitrogen-bearing minerals in this meteorite. No nitrogen isotope determinations have been reported for the intermediate EH5 chondrites.

Aubrites are achondritic meteorites, mainly coarse-grained breccias. The importance of this group of eleven meteorites lies in its apparent close relationship with E-chondrites, and the implications such a relationship has for their origin. Major, minor and trace element composition of minerals in aubrites show great similarity to EL6's, hence it has been suggested that aubrites formed from EL6-type precursor material by igneous differentiation (WATTERS and PRINZ, 1979; BISWAS *et al.*, 1980; WOLF *et al.*, 1983). Alternatively, they may be primitive nebular condensates, the extensive recrystallization textures are a result of either burial at depth within a parent body (SEARS, 1980), or surface-location and melting by an external heat source (WASSON and WAI, 1970). A separate parent body altogether has also been proposed (BRETT and KEIL, 1985).

The carbon content of aubrites is much lower than that of enstatite chondrites, with abundances ranging from 0.02–0.14 wt% (TROFIMOV, 1950; BELSKY and KAPLAN, 1970; GIBSON *et al.*, 1971). At present, the exact nature of carbon in aubrites is unclear, but may be mostly graphite (DODD, 1981). Literature $\delta^{13}C$ values for enstatite achondrites are completely different from enstatite chondrites and range from −27.5 to −23.9‰ (BELSKY and KAPLAN, 1970; YANG and EPSTEIN, 1983). Nitrogen abundances in aubrites are much lower than those in enstatite chondrites, ranging from 10–100 ppm (GIBSON *et al.*, 1971; INJERD and KAPLAN, 1974; KUNG and CLAYTON, 1978; CARR and PILLINGER, 1984). Data for enstatite achondrites are limited to one measurement by KUNG and CLAYTON (1978) on Norton County ($\delta^{15}N$ = +4.4‰) and a study by CARR and PILLINGER (1984) of Bustee where the $\delta^{15}N$ of an osbornite crystal extracted from a CaS nodule was −24‰.

SAMPLES AND ANALYSIS

Ten enstatite chondrites (four EH4, one EH5 and five EL6), plus one aubrite were analyzed for carbon and nitrogen abundances and isotopic compositions; separate aliquots of material

were used for each determination. The meteorites analyzed, and the source from which they were obtained are listed in Table 1. Each sample, an interior chip, *ca.* 200 mg, was crushed to a grain size approximately 50 μm. Aliquots for analysis were drawn from this reservoir: between 30–60 mg per carbon experiment, and typically 0.5 mg for nitrogen. The sample of Abee was derived from clast (1,1) (MARTI, 1983).

Stepped heating in an atmosphere of oxygen was used to obtain both carbon and nitrogen data. Additionally, separate aliquots of the meteorites were treated to stepped pyrolysis for nitrogen measurements. Full experimental details for the carbon extraction procedure are published in SWART *et al.* (1983a). Carbon was analyzed as carbon dioxide gas on a VG Micromass 602E mass spectrometer; errors for sample sizes greater than 2 μg of carbon are ±0.1‰. Smaller samples, measured *via* a gas-mixing technique, had correspondingly larger errors, *ca.* ±1‰. Blank levels for the carbon extraction technique were <30 ng per half hour temperature step. Due to the low carbon concentration in the Bustee aubrite, measurement was made using a high sensitivity static mass spectrometer (CARR *et al.*, 1986), which allows amounts of carbon down to 2 ng to be analyzed with a precision of ±1‰. Nitrogen isotope analyses were performed on a double collector, static mass spectrometer, with errors of ±2‰ on the minimum sample size of 2 ng. Blanks for nitrogen analyses were <2 ng per temperature step. Data are reported relative to international standards: PDB for carbon and AIR for nitrogen.

The use of stepped combustion in certain favourable cases allows identification of discrete components in meteorites. In order to assess the suitability of the technique for analysis of enstatite chondrites, individual materials were treated to the standard analytical procedures. Figure 1 shows the combustion profiles of (a) carbon dioxide released from graphite and cohenite and (b) nitrogen released from cohenite, osbornite and sinoite. It is accepted that a multi-component mixture may combust differently from isolated compounds, and that there may be grain-size and catalytic effects plus degrees of mineral protection, etc. However, from Fig. 1a, it seems conceivable that carbon as cohenite could be resolvable from carbon as graphite. Indeed, a mixture of the two reference compounds affords separate peaks in the combustion profile (I. A. FRANCHI, pers. commun.). The nitrogen content of cohenite is low (<100 ppm, Fig. 1b) and would probably be masked by the release of nitrogen from nitride minerals. Osbornite seems to combust at slightly lower temperature than sinoite, therefore partial resolution of these components by stepped combustion might be feasible if they possess different isotopic signatures.

RESULTS AND DISCUSSION

(a) Carbon

Carbon concentration and isotopic data from stepped combustion of ten enstatite chondrites are listed in Table 2; a summary of whole rock data is given in Table 3. The abundance of carbon in EH4's ranges from 0.15 wt% to 0.42 wt%, with the lowest value registered for South Oman, which extends the known scale. For EL6's, the samples measured herein have carbon abundances between 0.29 wt% and 0.70 wt%, within the published range. St. Mark's, the single EH5 measured, with a carbon abundance of 0.53 wt%, falls above the end of the range for EH4's, but within that of the EL6 group. Hence the highest petrographic grade samples show the widest variation in abundance of the volatile element. The whole-rock $\delta^{13}C$ values for EH4 samples agree well with literature data, except for South Oman which is again unusual: $\delta^{13}C = -28.5$‰, over 15‰ lighter than most other enstatite chondrites, which have a total range in $\delta^{13}C$ values of 10‰. This may be due to either the presence of terrestrial contaminants or an indigenous ^{12}C-enriched component (see below). The $\delta^{13}C$ measurements for EL6's are in less good agreement with previously published observations, and are generally higher than for EH4 meteorites, as suggested previously (DEINES and WICKMAN, 1985). The major carbonaceous constituents of E-chondrites are graphite and carbon in metal, both known to combust at elevated temperatures (Fig. 1a). In order to compare the $\delta^{13}C$ values of these species in different meteorite samples, only the carbon combusting at temperatures > 700°C is considered (Fig. 2a, b). On this basis, South Oman has a $\delta^{13}C$ value very similar to other EH4 samples, although the meteorite is deficient in that carbon which combusts at

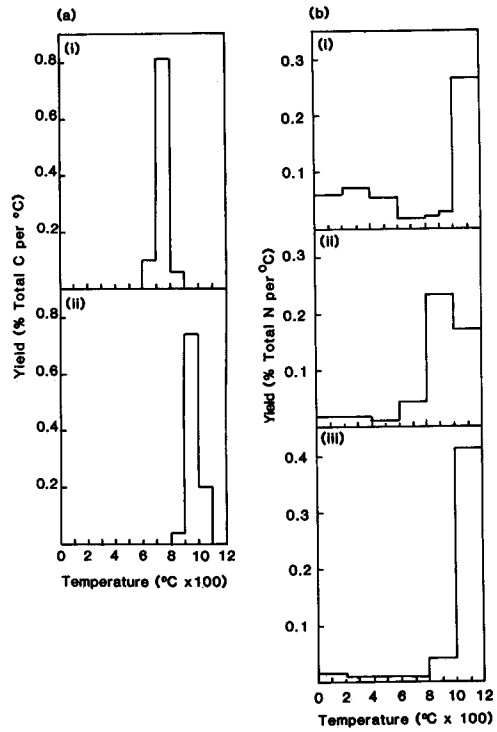

FIG. 1. (a) Carbon liberated by stepped combustion of (i) NBS-21 graphite and (ii) Canyon Diablo cohenite. (b) Nitrogen liberated by stepped combustion of (i) Canyon Diablo cohenite; (ii) artificial osbornite and (iii) artificial sinoite.

Table 1: Samples analyzed and their source

Sample	Type	Source+
Abee*	EH4	UCSD (1,1,02,0)
Indarch*	EH4	BM 86948
Kota-Kota	EH4	BM 1905,355
South Oman	EH4	SOA 1028
St Mark's*	EH5	BM 1970,339
Atlanta	EL6	BM 1959,1001
Daniel's Kuil*	EL6	BM 42507
Khairpur*	EL6	BM 51366
North West Forrest (1971)	EL6	WAM 13194
Yilmia	EL6	BM 1972,132
Bustee*	Aub	BM 32100

+Abbreviations: BM - British Museum (Natural History), London; SOA - Dr S. O. Agrell, Cambridge; UCSD -Dr K. Marti, Univ. of California at San Diego (consortium study, clast 1,1,02,0); WAM - Western Australian Museum, Perth.

*Observed fall

Table 2: Carbon data from stepped combustion of 10 enstatite chondrites

Sample	Abee		Indarch		Kota-Kota		South Oman		St. Mark's	
Wt(mg)	54.61		35.94		42.83		53.45		51.03	
Temp. °C	Yield ppm C	$\delta^{13}C$ ‰	Yield ppm C	$\delta^{13}C$ ‰	Yield ppm C	$\delta^{13}C$ ‰	Yield ppm C	$\delta^{13}C$ ‰	Yield ppm C	$\delta^{13}C$ ‰
200	22	-26.3	75	-22.9	152	-17.7	23	-32.4	25	-25.4
300	159	-30.5	120	-31.4	196	-35.5	229	-28.6	382	-29.2
400	178	-28.4	362	-22.9	339	-18.9	187	-24.9	435	-23.8
450	238	-19.7	242	-22.9	-	-	-	-	197	-19.5
500	105	-18.3	187	-19.0	361	-9.5	530	-46.6	137	-17.2
550	962	-17.0	-	-	-	-	-	-	-	-
600	46	-18.1	136	-15.9	682	-8.4	176	-16.0	192	-20.6
650	38	-20.4	-	-	-	-	-	-	-	-
700	99	-13.3	314	-8.7	465	-7.1	91	-10.1	151	-11.0
800	139	-10.7	1027	-8.2	255	-6.5	51	-12.6	1000	-6.3
900	465	-9.2	759	-7.6	418	-5.9	154	-7.1	1638	-7.1
1000	889	-8.0	520	-7.5	370	-6.2	38	-10.1	915	-4.4
1100	470	-5.1	-	-	67	-1.4	-	-	221	-3.7
1200	42	-7.6	65	-19.1	11	+3.7	27	-12.7	27	-17.0
Total	4212	-13.4	3807	-12.3	3316	-10.6	1506	-28.5	5320	-10.7

Sample	Atlanta		Daniel's Kuil		Khairpur		NW. Forrest		Yilmia	
Wt(mg)	49.99		63.56		58.41		37.36		44.85	
Temp. °C	Yield ppm C	$\delta^{13}C$ ‰	Yield ppm C	$\delta^{13}C$ ‰	Yield ppm C	$\delta^{13}C$ ‰	Yield ppm C	$\delta^{13}C$ ‰	Yield ppm C	$\delta^{13}C$ ‰
200	83	-17.8	48	-29.2	44	-31.7	116	-12.8	6	nm
300	280	-26.3	604	-28.4	967	-29.3	391	-17.7	35	-31.2
400	207	-20.0	325	-24.7	329	-27.2	311	-22.4	31	-28.8
450	166	-19.0	64	-25.1	156	-26.7	-	-	-	-
500	207	-41.6	48	-28.1	112	-26.3	115	-13.2	169	-23.5
600	105	-8.5	120	-27.3	170	-31.8	264	-6.7	272	-5.1
700	226	-3.3	24	-19.1	189	-5.6	809	-3.7	380	-3.6
800	1238	-3.5	77	-0.4	502	-2.5	3110	-1.7	358	-4.9
900	1310	-5.1	300	+3.3	750	-4.1	1172	-0.6	1130	-7.0
1000	1296	-5.3	860	+6.5	523	-0.6	77	-7.5	1089	-7.2
1100	992	-5.3	286	+1.6	169	-0.7	-	-	281	-4.8
1200	258	-3.3	94	-1.6	108	-1.1	18	-16.8	27	-13.3
Total	6368	-8.0	2850	-9.3	4019	-14.1	7003	-4.1	3778	-7.4

high temperatures. The isotopic composition of >700°C carbon in EH4 chondrites exhibits only a small spread in $\delta^{13}C$ from −9.1 to −5.8‰, while that in EL6 meteorites is more variable, and generally isotopically heavier: $\delta^{13}C$ from −6.7 to +4.2‰. The high abundance of >700°C carbon and the relative ^{13}C-enrichment for this material explains the higher bulk $\delta^{13}C$ values discussed earlier for EL6 chondrites.

Notwithstanding problems of sample heterogeneity, it seems that different meteorites contain widely variable amounts of carbon burning above 700°C, ranging from 0.03 to 0.24 wt% in EH4's and 0.16 to 0.51 wt% in EL6's. Although there is some overlap in the >700°C carbon content between the two meteorite groups, it can be seen that the highest carbon contents are observed in the EL6 samples. Petrographic observations indicate that EL6 chondrites contain more graphite than EH4 specimens, although KEIL (1968) also sug-

Table 3: Summary of $\delta^{13}C$ and carbon concentration, $\delta^{15}N$ and nitrogen concentration from stepped combustion and stepped pyrolysis extractions of enstatite chondrites.

Sample	C Combustion Wt(mg)	$\delta^{13}C$ ‰	[C] ppm	N Combustion Wt(mg)	$\delta^{15}N$ ‰	[N] ppm	N Pyrolysis Wt(mg)	$\delta^{15}N$ ‰	[N] ppm
Abee	54.61	-13.4	4212	0.567	-29.2	531	0.407	-12.7	973
							0.406	-7.0	481
							1.672	-15.7	220
Indarch	35.94	-12.3	3807	0.764	-22.1	380	2.798	-30.5	897
							1.679	-24.1	428
Kota-Kota	42.83	-10.6	3316	0.766	-19.7	310	0.940	-26.1	534
South Oman	53.45	-28.5	1506	0.406	-27.4	288	0.457	+0.6	787
				4.545	-16.6	147			
St. Mark's	51.03	-10.7	5320	0.525	-6.3	152	0.557	-12.2	195
Atlanta	49.99	-8.0	6368	0.435	-24.9	210	0.500	-10.6	326
Daniel's Kuil	63.56	-9.3	2850	0.881	-15.2	133	2.778	-31.0	475
Khairpur	58.41	-14.1	4019	1.060	-24.3	252	3.492	-47.4	946
N.W. Forrest	37.36	-4.1	7003	1.284	-23.4	200	2.385	-44.6	490
Yilmia	44.85	-7.4	3778	0.756	-22.7	193	0.523	-24.2	215

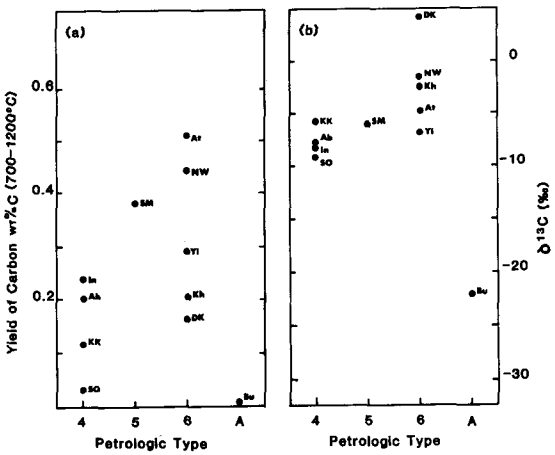

FIG. 2. Variation of (a) carbon yield and (b) $\delta^{13}C$ with petrologic type for carbon combusting at temperatures greater than 700°C (*i.e.* predominantly graphite). A—aubrite; Ab—Abee; At—Atlanta; Bu—Bustee; DK—Daniel's Kuil; In—Indarch; Kh—Khairpur; KK—Kota-Kota; NW—North West Forrest (1971); SM—St. Mark's; SO—South Oman; Yi—Yilmia.

gested that a proportion of the carbon may be present as either cohenite or in kamacite.

Graphite is known to be a product of thermal metamorphism in terrestrial rocks, however it is difficult to see how metamorphism could be responsible for the graphite distribution in enstatite chondrites, where the highest grade rocks have an increased abundance of carbon. In the case of ordinary chondrites, the higher petrologic types of each chemical group contain less carbon (graphite) than samples from the same group which are thought to have experienced lower degrees of metamorphism. It is also interesting to note that graphite (in association with magnetite) in L-group chondrites has $\delta^{13}C$ some 7–10‰ heavier than in H-group meteorites (GRADY *et al.*, 1982). For several reasons the H- and L-group ordinary chondrites are thought to have formed on separate parent bodies; the large difference in isotopic composition of graphitic carbon in the two groups supports this. By analogy with ordinary chondrites, the difference in $\delta^{13}C$ of graphite in H- and L-group enstatite chondrites may also be suggesting separate parent bodies, rather than metamorphic effects. Subtle variations in isotopic composition within EH4 and EL6 samples are not yet understood, but the observation that within each type there is no apparent relationship of $\delta^{13}C$ with carbon content would seem to exclude fractionation by thermal metamorphism.

Figure 3 shows the carbon release profiles for all the enstatite chondrites analyzed by stepped combustion. In no case is the major release of carbon as clearly defined as that of the reference graphite material (Fig. 1a). Mostly, the maximum amount of carbon burns at 800–900°C, a temperature some 100°C higher than standard graphite. Only Indarch (EH4) and North West Forrest (EL6) show carbon which burns sharply at a lower temperature than 800°C. In contrast, for Atlanta carbon seems to burn over a very extended temperature range (700–1100°C). In E-chondrites graphite is presumably present with a range of grain sizes, in sites "protected" by minerals resistant to oxidation or finely dispersed in metal. All of these effects will result in extended or variable combustion intervals. Similar problems were encountered with ureilites (GRADY *et al.*, 1985), but were not seen in studies of the graphite-

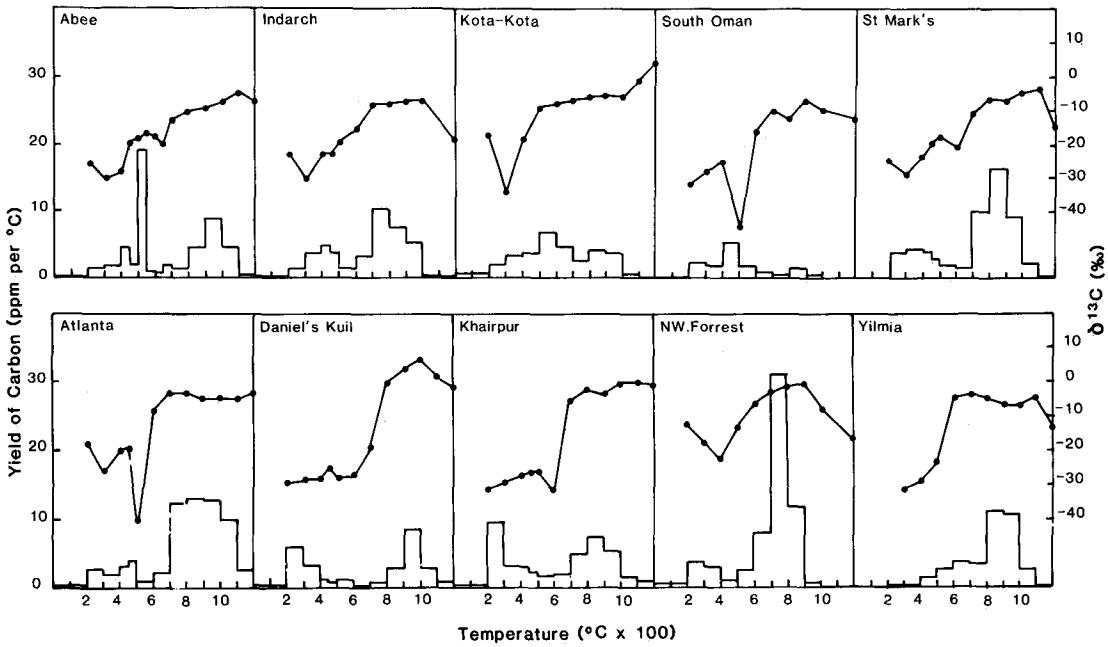

FIG. 3. Abundance and isotopic composition of carbon liberated by stepped combustion of ten E-chondrites, (——) yield; (— ● —) $\delta^{13}C$.

rich winonaites (GRADY and PILLINGER, 1986). The presence of cohenite or carbon in solid-solution in iron metal may also contribute to the nature of the combustion profile. Several of the chondrites appear to possess varying amounts of carbon which combusts at temperatures as low as 500–600°C, with $\delta^{13}C$ similar to that of carbon at higher temperatures. This material may be amorphous carbon or highly disordered graphite, since crystallinity can also have an effect on combustion temperature. Over the temperature range 600 to 1200°C there is a tendency for $\delta^{13}C$ to increase, although Atlanta and Yilmia are notable exceptions. This may be interpreted as (i) a mixing between low temperature, isotopically light material (indigenous or contaminant) and graphite (considered unlikely, since organic material is usually combustible below 600°C); (ii) a kinetic isotope fractionation during combustion of graphite (an effect which was not observed during combustion of the reference graphite); (iii) combustion of two different graphite components with slightly different isotopic compositions; (iv) combustion at high temperatures of small amounts of ^{13}C-enriched interstellar grain material, or (v) simultaneous combustion of carbon in solid-solution in metal, which combusts over a broad temperature range with graphite (GRADY and PILLINGER, 1986). Graphite is known to occur in enstatite chondrites in more than one site, within silicate grains and in association with metal and carbides. It is feasible that at elevated temperatures (*i.e.* during metamorphism) graphite in silicates has taken part in reduction reactions, leading to loss of carbon as CO. The result of reactions of this type would be the enrichment of residual carbon in ^{13}C (BOTTINGA, 1969), and thus graphite with different isotopic compositions may be present. Alternatively, it is known that enstatite chondrites contain small amounts of isotopically anomalous carbon; acid-resistant residues showing $\delta^{13}C$ values of up to +160‰ in Indarch (CARR *et al.*, 1984) and +389‰ in Qingzhen (WRIGHT *et al.*, 1986). The demineralization/wet-chemical oxidation techniques used in these studies allowed the isotopically anomalous carbon components to be readily resolvable from graphite using stepped combustion. Kota-Kota, which contains only relatively small amounts of graphite, has a $\delta^{13}C$ value of +3.7‰ at 1200°C, which, apart from the value of +6.5‰ measured in Daniel's Kuil at 1000°C, represents the heaviest carbon isotopic composition measured in this study. The small amounts of graphite in Kota-Kota may have allowed the putative ^{13}C-rich component to be detected more readily. However, it should be noted that carbon analyses of St Sauveur (EH5) and Yilmia (EL6) have failed to uncover any isotopically anomalous carbon components (WRIGHT *et al.*, 1986).

The presence of cohenite would also be indicated by a release of carbon at higher temperatures. As outlined by DEINES and WICKMAN (1975), there are two opposing constraints on cohenite formation. Whilst it seems that metal in enstatite chondrites (generally 6 to 8% nickel, as measured by electron probe; KEIL, 1968) is a suitable medium for cohenite to form (as predicted by BRETT, 1967), high concentrations of silicon in E-chondrite metal are likely to destabilize any cohenite, resulting in graphite formation. Notwithstanding these two opposing effects, cohenite has been positively identified in Abee (HERNDON and RUDEE, 1978). From Fig. 1a, it may be seen that a standard sample of cohenite combusts at a higher temperature than graphite. None of the thermal release profiles for enstatite chondrites shows evidence for a distinct peak which could be ascribed to cohenite, but interestingly, in six of the samples (Indarch, South Oman, St. Mark's, Daniel's Kuil, North West Forrest and Yilmia, but unfortunately not Abee), carbon which is isotopically lighter than the main graphite release is found in the highest temperature step(s), which might be taken as circumstantial evidence for the existence of cohenite. More definitive evidence might be expected from the analysis of separated metal components.

In all samples except Yilmia, significant amounts of carbon are observed at low temperatures (200–600°C). Indeed, for Abee, Kota-Kota, South Oman and Khairpur, the maximum carbon release is encountered before 600°C. It is tempting to dismiss this carbon as terrestrial contamination, and for the large amounts of carbon combustible at temperatures between 200–300°C, with $\delta^{13}C$ *ca.* −30‰, seen in Daniel's Kuil (fell 1868) and Khairpur (fell 1873) this is probably a reasonable interpretation. Yet meteorite "finds" (*e.g.* Yilmia, found 1969), which would be expected to be contaminated to a greater extent, have only small quantities of low temperature carbon, and are not consistent with such a hypothesis. All the EH4's and some EL6's show evidence for a component of isotopically light carbon burning somewhere between 300 and 500°C—this is most clearly seen in South Oman ($\delta^{13}C$ = −46.6‰, at 400–500°C) and Atlanta ($\delta^{13}C$ = −41.6‰ at 450–500°C), where there is a corresponding peak in carbon release. The $\delta^{13}C$ values of < -40‰ are amongst the lowest reported for meteorites (SWART *et al.*, 1983a), and are not expected from normal terrestrial contaminants, whereas carbon-bearing components enriched in ^{12}C have been recognized as the host phase of Xe (HL) in the Allende and Murchison carbonaceous chondrites (SWART *et al.*, 1982, 1983b). Since Xe (HL) has been identified as a noble gas component in some enstatite chondrites (CRABB and ANDERS, 1981, 1982) (although unfortunately not in South Oman), it is likely that a similar host phase is also present. This interpretation is supported by nitrogen isotope data (see following section). In the investigation described here, only whole-rock material was step combusted, using fairly large temperature increments of 100°C. However, the $\delta^{13}C$ values are lower than those measured for acid-resistant residues from Murchison and Allende, hence although resolution of the noble gas host phase from "normal" carbon is still likely to be a problem with E-chondrites, the more extreme $\delta^{13}C$ values of −41.6‰ (Atlanta) and −46.6‰ (South Oman) suggest the Xe (HL) carrier

may occur in higher concentrations and may be isolated more readily.

The Xe (HL) host is not the only low temperature indigenous carbon phase of interest in enstatite chondrites. Stepped combustion of Abee indicates the presence of another discrete carbonaceous component, burning sharply between 500 and 550°C, with $\delta^{13}C$ *ca.* −17.5‰. The release temperature and isotopic composition do not suggest terrestrial contamination (SWART *et al.*, 1983a), and indeed Abee is a recent fall. This component may be indigenous organic material, similar in $\delta^{13}C$ value to the insoluble macromolecular material in carbonaceous chondrites (BELSKY and KAPLAN, 1970; SMITH and KAPLAN, 1970; ROBERT and EPSTEIN, 1982; GILMOUR *et al.*, 1985), or could result from decrepitation of a carbonate produced by weathering (DEINES and WICKMAN, 1985). An intriguing possibility, however, is that the constituent may be "mysterite" (LAUL *et al.*, 1973; GANAPATHY and LARIMER, 1980). An important question with regard to the sharp carbon release is whether or not it is associated with isotopically anomalous nitrogen (see below).

Carbon occurs in aubrites in much lower concentrations: at 0.03 wt% *ca.* an order of magnitude lower than that in E-chondrites. Interpretations of carbon data for the aubrite Bustee (Table 4) must also be included in any scheme which inter-relates the various groups of enstatite meteorites. Examination of carbon data (Table 4 and Fig. 4) indicates that there is a peak release of carbon at high temperature (800–1000°C), possibly graphite, with a $\delta^{13}C$ between −24 and −20‰. Above 1000°C, isotopically lighter carbon is released ($\delta^{13}C$ = −30‰), which may be from cohenite or carbon in metal. The difference between the two isotopic compositions is in the correct sense for the graphite to be a product of exsolution (SEARS, 1980). Carbon liberated at all temperatures above 600°C is more enriched in ^{12}C compared to that of enstatite chondrites, which is the reverse of that which might be expected if aubrites were derived from enstatite chondrites by a simple volatilization process. On this basis, it is most likely that aubrites derive from a different parent body from the enstatite chondrites. The most surprising feature in the stepped combustion carbon release profile of Bustee is the large peak at 500°C which may be either organic, or from decrepitation of a carbonate, although whether indigenous to the meteorite or contaminant is not certain. The isotopic composition ($\delta^{13}C$ = −24‰) is unusual for a terrestrial carbonate, but within the accepted range of terrestrial organic contaminants.

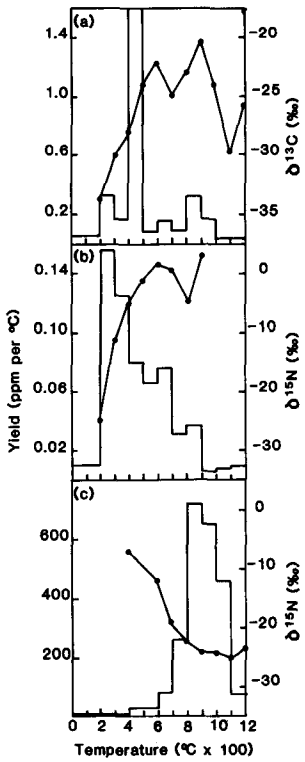

FIG. 4. Stepped extraction profiles for the Bustee aubrite (a) carbon whole-rock combustion; (b) nitrogen pyrolysis of residue after removal of osbornite by hand-picking and (c) nitrogen pyrolysis of an osbornite separate, (——) yield; (— ● —) isotopic composition.

Table 4: Carbon and nitrogen data from Bustee aubrite

Sample Temp. °C	C whole rock* Yield ppm	$\delta^{13}C$ ‰	N TiN removed* Yield ppm	$\delta^{15}N$ ‰	N TiN only* Yield %	$\delta^{15}N$ ‰
200	(12)	(-34)	(2.0)	(-25)	-	-
300	(33)	(-30)	(15.6)	(-11)	-	-
400	(17)	(-28)	(12.5)	(-5)	(0.3)	(-7)
500	160	-24	8.0	-1	-	-
600	8	-22	6.6	+2	0.6	-12
700	16	-25	7.6	+1	0.8	-19
800	9	-23	3.1	-5	2.6	-22
900	32	-20	3.7	+3	7.2	-24
1000	17	-24	0.6	nm	6.5	-24
1100	4	-30	0.8	nm	4.6	-25
1200	4	-26	1.0	nm	0.7	-24
Total+	250	-24	31.4	0	23.0	-24

*Carbon data were acquired by stepped combustion, nitrogen by stepped pyrolysis.
+Totals for carbon and nitrogen exclude data up to 400°C (shown in brackets). Sample weights are 10.09 mg for carbon analysis and 5.12 mg for nitrogen; the weight of the osbornite separate was uncertain, but probably < 1μg.

(b) Nitrogen

Nitrogen data obtained by stepped combustions and stepped pyrolyses of the enstatite chondrites are shown in Tables 5 and 6, respectively; a summary of whole-rock data is given in Table 3. Also included in Table 3 are data from replicate experiments (two repeat stepped pyrolyses of Abee, a second pyrolysis extraction of Indarch and a second combustion of South Oman). For pyrolysis extractions, data for the 1200°C step is a composite of results from a pyrolysis followed by a combustion at 1200°C. In most cases, $\delta^{15}N$ values of the combustion step were very similar to those obtained from the preceding (*i.e.* 1200°C) pyrolysis step, and this final step accounts for the largest proportion of the nitrogen. In most cases, the samples melted, indicating that nitrogen release was complete. The py-

Table 5: Nitrogen data from stepped combustion of 10 enstatite chondrites

Sample	Abee		Indarch		Kota-Kota		South Oman (1)		South Oman (2)		St Mark's	
Wt(mg)	0.567		0.764		0.766		0.406		4.545		0.525	
Temp. °C	Yield ppm N	$\delta^{15}N$ ‰	Yield ppm N	$\delta^{15}N$ ‰	Yield ppm N	$\delta^{15}N$ ‰	Yield ppm N	$\delta^{15}N$ ‰	Yield ppm N	$\delta^{15}N$ ‰	Yield ppm N	$\delta^{15}N$ ‰
200	30	-64.4	3	-17.4	54	-21.0	10	+11.5	-	-	2	-2.8
300	185	-62.1	23	-1.6	77	-21.4	68	-72.4	2	-2.5	5	+2.1
400	44	+25.5	19	-2.4	42	-14.1	35	-15.6	2	-1.6	11	+34.4
500	56	+57.4	23	-5.9	27	-7.8	20	+8.0	4	+9.9	-	-
600	23	-22.7	34	-11.4	17	-7.0	20	-15.9	8	+12.8	12	+13.6
700	17	-8.6	48	-13.0	11	-6.7	30	+0.6	20	+13.4	29	-10.2
800	19	-25.9	52	-26.6	8	-18.4	35	-12.0	8	+10.2	19	-8.5
900	30	-29.7	49	-29.2	9	-10.7	20	-24.8	25	+0.5	17	-6.8
1000	37	-18.0	38	-30.7	22	-31.3	11	nm	20	-16.9	16	-6.0
1100	36	-39.0	49	-36.1	21	-33.4	7	nm	17	-36.2	21	-14.7
1200	54	-41.3	42	-32.9	22	-31.9	32	-31.0	41	-48.5	20	-26.0
Total	531	-29.2	380	-22.1	310	-19.7	288	-27.4	147	-16.6	152	-6.3

Sample	Atlanta		Daniel's Kuil		Khairpur		NW. Forrest		Yilmia	
Wt(mg)	0.435		0.881		1.060		1.284		0.756	
Temp. °C	Yield ppm N	$\delta^{15}N$ ‰	Yield ppm N	$\delta^{15}N$ ‰	Yield ppm N	$\delta^{15}N$ ‰	Yield ppm N	$\delta^{15}N$ ‰	Yield ppm N	$\delta^{15}N$ ‰
200	13	-4.2	25	-10.2	43	-19.5	24	-4.7	13	nm
300	-	-	15	-0.8	28	-2.6	40	-2.9	-	-
400	60	-51.5	10	+11.9	17	-3.1	30	-2.4	39	-23.9
500	-	-	9	+11.4	22	-7.1	7	-14.5	-	-
600	29	-73.3	6	+4.5	16	-5.6	5	-0.5	19	-19.8
700	-	-	13	-10.4	16	-10.9	2	nm	-	-
800	24	+1.4	12	-9.6	9	-11.7	1	nm	42	-16.0
900	-	-	5	-35.4	10	-31.1	2	nm	-	-
1000	42	-11.7	5	-20.5	18	-39.8	7	-23.2	43	-19.1
1100	31	+19.1	13	-38.4	28	-48.5	19	-43.9	19	-29.9
1200	11	-5.7	20	-48.4	45	-50.1	63	-49.5	18	-39.6
Total	210	-24.9	133	-15.2	252	-24.3	200	-23.4	193	-22.7

rolysis experiments (including a 1200°C combustion) appear to release consistently more nitrogen than the combustion (Table 3), an artefact of the experimental technique used at the time arising from gas transfer difficulties, but known not to result in isotopic fractionation (LEWIS *et al.*, 1983).

The repeat experiments on Abee, Indarch and South Oman show that there is a wide variation in nitrogen content within a sample, *e.g.* the nitrogen concentration of Abee as determined by pyrolysis ranges from 220 to 973 ppm. The Abee data are for a single clast; THIEMENS and CLAYTON (1983) found differences in nitrogen content of separate clasts from Abee (254 to 853 ppm), but only a small variation between repeated analyses on the same clast. However, it should be stressed that Thiemens and Clayton used gram-sized samples, as opposed to the milligram and submilligram aliquots (taken from 0.2 g) analyzed and reported herein. MOORE *et al.* (1969) also found large variations in nitrogen contents of enstatite chondrites (241 to 343 ppm ion Abee; 582 to 710 ppm in Hvittis). Necessarily, the effects of using small samples and a heterogeneous distribution of nitrogen-bearing components must be considered when interpreting the data. Regardless of these problems, the data in Table 3 suggest that enstatite chondrites contain 100 to 1000 ppm nitrogen, with no systematic trend with petrologic type, in agreement with the range of 50 to 800 ppm reported by MOORE *et al.* (1969). Whole-rock $\delta^{15}N$ values range from −47.4 to +0.6‰, with variations of up to 11‰ from repeat experiments of the same sample. There also appears to be no relationship between $\delta^{15}N$ and petrologic type.

Stepped combustion profiles of all the enstatite chondrites examined are shown in Fig. 5; stepped pyrolysis data are in Fig. 6. In an attempt to detect any relationship between carbon- and nitrogen-bearing components, plots of C/N ratio *versus* temperature for the stepped combustion data of enstatite chondrites are shown in Fig. 7. In all cases, a maximum in the C/N ratio is observed between 800–1000°C, lending support to the conclusion that graphite (a nitrogen-free component) is burning in this temperature region. The C/N ratio over this temperature range is lowest (*ca.* 10) in South Oman, while in North West Forrest it reaches a maximum of 3600 at 800°C. This is consistent with South Oman and North West Forrest having low and high graphite contents, respectively. In general, the difference in C/N ratio over this temperature interval may reflect the variable contribution of carbon and nitrogen dissolved in metal (low C/N ratio) and graphite (high C/N ratio). The lower C/N ratios observed in EH4 meteorites argues for a higher proportion of metal-sited carbon and nitrogen in these samples than in EL6's.

At 1200°C, the C/N ratio drops dramatically to values between 0.25 and 5 for all meteorites studied except Atlanta. This may correspond to a release of nitrogen from a refractory component which is presumably carbon-free. Candidates for such a material include (i) nitride minerals, *e.g.* osbornite and sinoite; (ii) nitrogen substituted for oxygen within silicate lattices, or (iii) nitrogen in solid-solution in iron metal. The absolute amounts of nitrogen released at 1200°C during stepped combustion range from 10 to 50 ppm, increasing to

Table 6: Nitrogen data from stepped combustion of 10 enstatite chondrites

Sample	Abee (1)		Abee (2)		Abee (3)		Indarch (1)		Indarch (2)	
wt (mg)	0.407		0.406		1.672		2.798		1.679	
Temp. °C	Yield ppm N	$\delta^{15}N$ ‰	Yield ppm N	$\delta^{15}N$ ‰	Yield ppm N	$\delta^{15}N$ ‰	Yield ppm N	$\delta^{15}N$ ‰	Yield ppm N	$\delta^{15}N$ ‰
200	7	nm	-	-	5	-4.0	13	-23.0	-	-
300	40	-12.2	-	-	7	+1.0	19	-12.0	-	-
400	33	-15.0	92	-21.4	5	+3.0	28	-13.0	2	+63.0
450	-	-	-	-	1	+13.0	-	-	-	-
500	13	+269.4	-	-	2	-8.0	57	-5.0	1	+78.0
550	-	-	-	-	7	+350.0	-	-	13	-2.0
600	29	-2.6	29	+230.2	5	-10.0	54	+2.0	14	-3.0
650	-	-	-	-	-	-	-	-	15	-5.0
700	95	-19.2	27	+124.0	12	-17.0	81	-26.0	34	-9.0
750	-	-	-	-	-	-	-	-	51	-24.0
800	83	-5.4	27	-19.4	7	-28.0	140	-37.0	58	-27.0
850	-	-	-	-	-	-	-	-	43	-29.0
900	59	+4.1	37	-32.4	7	-31.0	103	-33.0	25	-25.0
1000	161	-18.5	46	-11.4	8	-34.0	95	-36.0	67	-35.0
1100	190	-27.4	44	-34.0	42	-34.0	65	-39.0	51	-27.0
1200	263	-17.3	179	-41.9	112	-32.0	241	-40.0	54	-30.0
Total	973	-12.7	481	-7.0	220	-15.7	896	-30.5	428	-24.1

Sample	Kota-Kota		South Oman		St Mark's	
wt (mg)	0.940		0.457		0.577	
Temp. °C	Yield ppm N	$\delta^{15}N$ ‰	Yield ppm N	$\delta^{15}N$ ‰	Yield ppm N	$\delta^{15}N$ ‰
200	11	+9.0	-	-	6	+11.6
300	20	-10.0	-	-	-	-
400	20	-3.0	-	-	16	-3.0
500	34	-3.0	-	-	-	-
600	33	-10.0	213	+22.9	14	+0.9
700	38	-10.0	63	-25.5	11	+8.5
800	57	-20.0	42	+15.2	9	+16.5
900	88	-31.0	31	-27.2	9	+10.7
1000	82	-38.0	23	-13.4	13	+4.6
1100	51	-43.0	18	-41.9	13	+2.7
1200	99	-38.0	396	-3.8	104	-27.4
Total	533	-26.1	786	+0.6	195	-12.2

Sample	Atlanta		Daniel's Kuil		Khairpur		NW. Forrest		Yilmia	
wt (mg)	0.500		2.778		3.492		2.385		0.523	
Temp. °C	Yield ppm N	$\delta^{15}N$ ‰	Yield ppm N	$\delta^{15}N$ ‰	Yield ppm N	$\delta^{15}N$ ‰	Yield ppm N	$\delta^{15}N$ ‰	Yield ppm N	$\delta^{15}N$ ‰
200	0	nm	5	-10.0	10	-25.0	13	-19.0	2	-25.6
300	1	nm	27	-10.0	51	-31.0	18	-22.0	-	-
400	1	nm	53	-1.0	35	-22.0	41	-13.0	18	-10.0
500	4	nm	32	+0.5	13	-31.0	9	+6.0	-	-
600	6	nm	26	-0.1	22	-22.0	13	+5.0	13	+0.4
700	8	nm	24	-0.1	40	-19.0	16	-5.0	-	-
800	3	nm	19	-3.0	30	-18.0	9	-2.0	24	+2.1
900	7	nm	11	-7.0	25	-20.0	7	-22.0	-	-
1000	7	nm	13	-12.0	51	-18.0	15	-44.0	20	-6.6
1100	13	-1.3	19	-19.0	42	-22.0	14	-57.0	20	-2.2
1200	275	-11.0	245	-56.0	629	-60.0	335	-57.0	117	-41.3
Total	325	-10.6	474	-31.0	948	-47.4	490	-44.6	214	-24.2

ca. 100 ppm if the 1100°C step is included. Nitrogen released by combustion at 1200°C (excluding Atlanta) is isotopically light. The ^{15}N-depleted nitrogen release is more clearly seen in stepped pyrolysis data, where large amounts of the gas are liberated from all samples at 1200°C. In order to facilitate comparisons, $\delta^{15}N$ values recorded at 1200°C are plotted in Fig. 8. Ignoring Atlanta, which has a high C/N ratio (equal to 27) at 1200°C, it may be seen that EH4 chondrites exhibit a range in $\delta^{15}N$ of −41 to −4‰, while that in EL6 meteorites is from −60 to −40‰. St. Mark's (EH5), with $\delta^{15}N$ = −27 to −26‰ is within the spread of values for EH4's. Clearly, therefore, there is a dependence of $\delta^{15}N$ with petrologic type for nitrogen released at 1200°C.

A possible interpretation of the above observations is as follows: the relatively low nitrogen isotopic values noted for EL6 meteorites may arise from sinoite, which is believed on the basis of microscope studies to be more prevalent in EL6 than EH4 chondrites. The stepped combustion profiles of the three samples which contain the isotopically lightest nitrogen (Daniel's Kuil, Khairpur and North West Forrest) show that from 900–1200°C there is a monotonic increase in the amount of nitrogen released with increasing temperature. Data shown in Fig. 1b confirm that nitrogen from artificial

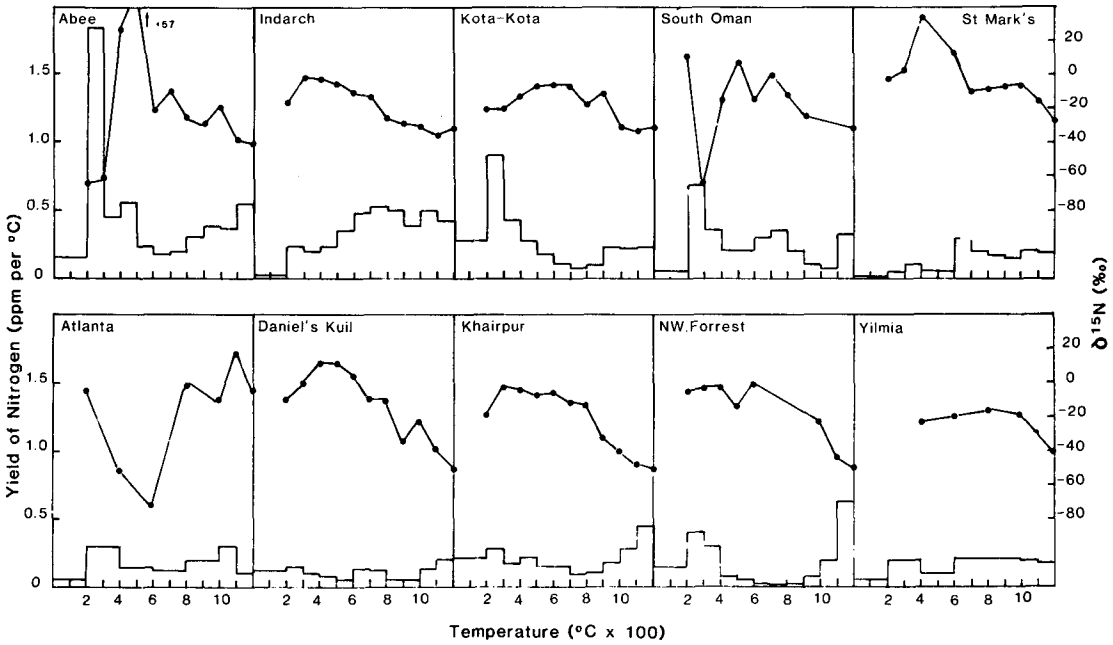

FIG. 5. Abundance and isotopic composition of nitrogen liberated by stepped combustion of ten enstatite chondrites, (——) yield; (— ● —) $\delta^{15}N$.

sinoite is released at the highest temperature (1200°C). For Yilmia and most of the EH4 samples, approximately equivalent amounts of nitrogen are released at each step between 900 and 1200°C. Recourse to Fig. 1b shows that artificial osbornite combusts from 800–1200°C, with a maximum in the release at a temperature of 1000°C. If these simplistic interpretations are correct, then Daniel's Kuil, Khairpur and North West Forrest contain the highest proportion of sinoite, whilst Yilmia and the EH samples appear to contain predominantly osbornite, and St. Mark's may contain osbornite only. However, this is only a simple model, since there are also other important sites of nitrogen-retention, particularly within the silicate lattice, which would also be expected to release nitrogen at elevated temperatures. It is possible that Atlanta contains neither

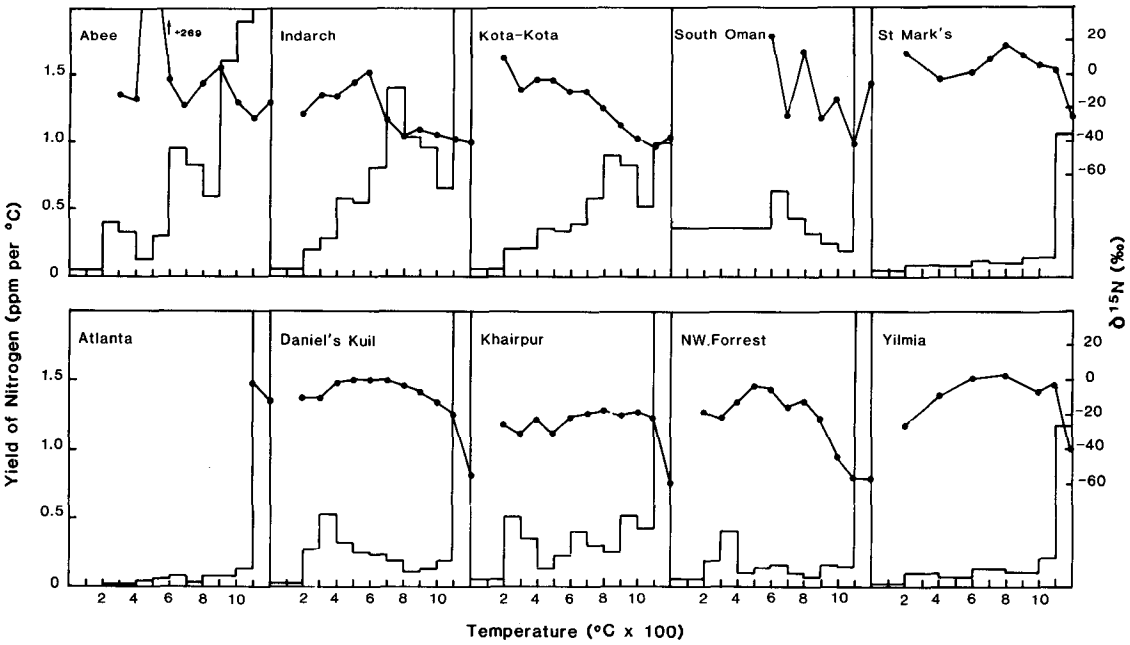

FIG. 6. Abundance and isotopic composition of nitrogen liberated by stepped pyrolysis of ten enstatite chondrites, (——) yield; (— ● —) $\delta^{15}N$.

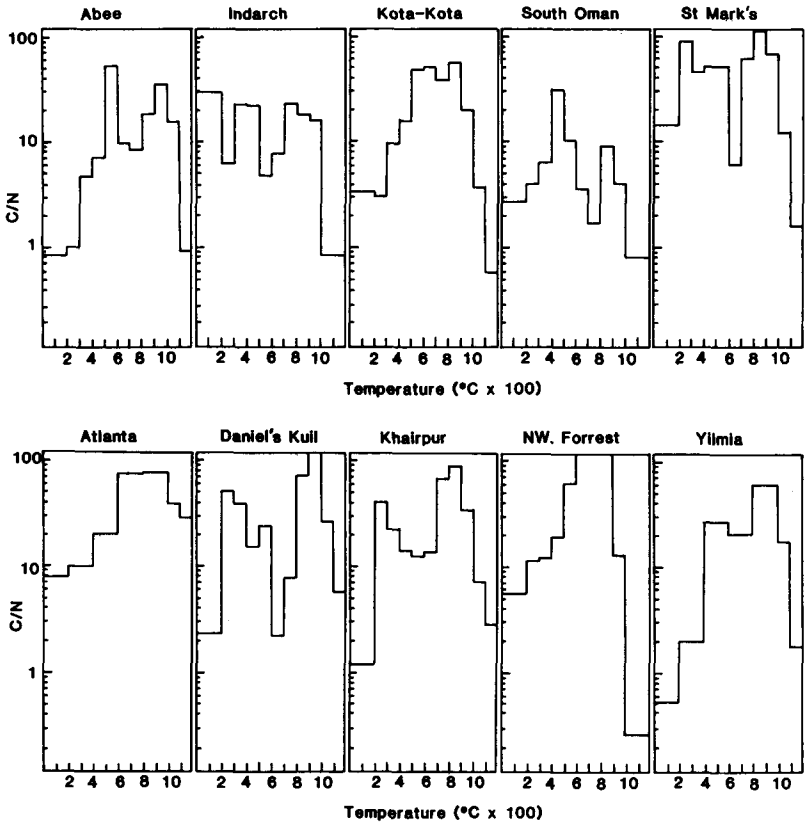

FIG. 7. Variation of C/N atomic ratio with temperature.

osbornite nor sinoite; the release of nitrogen ($\delta^{15}N$ = −11 to −6‰) above 1000°C and high C/N ratio of this material may suggest nitrogen associated with an unidentified carbon-containing component. However, there may be more than one high temperature component, since at least part of the high temperature nitrogen in Atlanta is isotopically heavy ($\delta^{15}N$ values reach +19‰ at 1100°C).

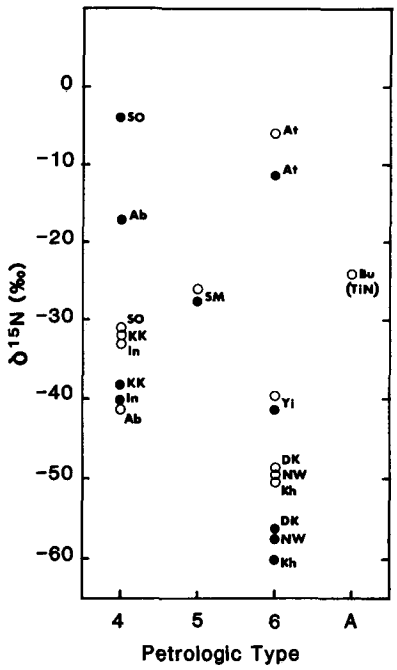

FIG. 8. Difference in $\delta^{15}N$ of nitrogen liberated by pyrolysis (O) and combustion (●) at 1200°C.

Nitrogen released by combustion at low temperatures (below 600°C) exhibits considerable variation in both release pattern and isotopic composition. Some samples (*e.g.* Abee, South Oman and Atlanta) show pronounced nitrogen releases or well-defined $\delta^{15}N$ maxima or minima, whereas other samples (*e.g.* Indarch, Khairpur) are comparitively featureless in both respects. There does not appear to be a dependence on petrologic grouping. In general terms, it is noteworthy that both isotopically light and heavy forms of nitrogen occur in enstatite chondrites at low temperatures (see Table 5). Isotopically light nitrogen is found in three samples: Abee ($\delta^{15}N$ = −64 to −62‰ at <300°C), South Oman ($\delta^{15}N$ = −72‰ from 200–300°C) and Atlanta ($\delta^{15}N$ = −73 to −51‰ from 200–600°C). Kota-Kota also has a marked nitrogen release at 300°C, but the $\delta^{15}N$ is no lower than −21‰. The isotopically light component must be heterogeneously distributed with respect to the sample sizes studied, since in the repeat combustion of South Oman, on a 4.5 mg sample (Table 3) the component was not detected. Isotopically light nitrogen, with $\delta^{15}N$ values down to −326‰, has been measured in demineralized and oxidized residues, particularly fractions < 1 μm

in size, prepared from carbonaceous chondrites (LEWIS *et al.*, 1983). However, in the case of the carbonaceous chondrites, release occurred at combustion temperatures between 400 and 600°C; ^{14}N-enriched nitrogen has never previously been detected at temperatures as low as those at which this component was released from whole-rock samples of enstatite chondrites. This could mean that two distinct isotopically light nitrogen components exist, or that the same component occurs in slightly different physical forms in carbonaceous and enstatite chondrites. For example, it might be protected by a higher abundance of organic polymer in the former, or in an even finer grain size ($\ll 1$ μm) in the latter. It should be reiterated at this point that both South Oman and Atlanta released isotopically light carbon (δ^{13}C *ca.* −46 to −41‰) in the temperature region 400–500°C, and that ^{13}C-depleted carbon was associated with ^{15}N-depleted nitrogen found in carbonaceous chondrites (LEWIS *et al.*, 1983; SWART *et al.*, 1983b). The temperature discrepancy between carbon and nitrogen release requires investigation by simultaneous δ^{13}C and δ^{15}N measurement during a single combustion experiment.

Another feature of the ^{14}N-enriched nitrogen component in carbonaceous chondrites is its stability in the absence of oxygen; under pyrolysis conditions, appropriate samples do not exhibit low δ^{15}N values until temperatures in excess of 900°C (LEWIS *et al.*, 1983; THIEMENS and CLAYTON, 1983). Low δ^{15}N values are not observed at low temperatures in the pyrolysis experiments on Abee, South Oman, Atlanta, or even Kota-Kota. A low temperature ^{15}N-enriched component in Abee (see below) is also more easily discernible, which suggests better resolution from an isotopically light nitrogen contribution. It is difficult to comment on the temperatures at which pyrolysis releases ^{15}N-depleted nitrogen due to the existence of sinoite and osbornite, but the pyrolysis of South Oman at least shows a low δ^{15}N at 700°C.

Nitrogen isotope analyses of demineralized and oxidized residues of the Indarch meteorite (CARR *et al.*, 1984) have shown that isotopically light nitrogen (δ^{15}N = −55‰) is liberated from 500–800°C. (Although Indarch is a meteorite for which whole-rock stepped combustion has revealed no evidence for isotopically light nitrogen, Table 6). The data from CARR *et al.* (1984) show two minima in δ^{15}N (at 600 and 800°C) with isotopically heavier nitrogen liberated in between. This has been interpreted as a broad release of isotopically light nitrogen with a superimposed co-release of isotopically "normal" nitrogen (possibly even a contaminant from acid treatment). This in turn suggests that the low temperature combustible ^{14}N-enriched nitrogen component(s) may be present in other enstatite chondrites, and lends support to the arguments that release temperature may be dependent on some physical characteristic such as site or grain size.

In the carbon isotope section above, it was seen that isotopically light carbon in low temperature combustion steps of bulk enstatite chondrites had the lowest δ^{13}C values measured, indicating that the Xe (HL) host phase might be abundant or easy to isolate. In contrast, however, nitrogen isotopic values for bulk samples or acid-resistant residues (CARR *et al.*, 1984) do not approach the minimum δ^{15}N encountered in carbonaceous chondrites. This would seem to imply that an isotopically normal, or heavier, nitrogen component decomposes in combustion experiments on enstatite chondrites at similar temperatures to that at which the Xe (HL) host combusts.

Two meteorites show evidence for small amounts of ^{15}N-enriched nitrogen at low temperatures during combustion experiments: Abee δ^{15}N = +57‰ at 500°C and St. Mark's δ^{15}N = +34‰ at 400°C, with a third, Daniel's Kuil giving a maximum δ^{15}N of +12‰ also at 400°C. There are grounds for believing that more than one nitrogen component is implicated, since pyrolysis experiments with the three samples give markedly different results. The isotopically "heavy" nitrogen recorded during stepped combustion of St. Mark's and Daniel's Kuil has properties consistent with organic nitrogen, since although this component generally combusts below 600°C, it is liberated during stepped pyrolysis over a broad range in temperatures from 200 to 1000°C.

The ^{15}N-enriched nitrogen component liberated from Abee is different in that it is released during pyrolysis at the same temperature as combustion. This material has been isolated in three separate pyrolyses of Abee, with δ^{15}N values of +269‰ (400–500°C); +230‰ (400–600°C) and +350‰ (500–550°C). Release of ^{15}N-enriched nitrogen is sharp, occupying only one or two consecutive temperature increments, with δ^{15}N between −20 and 0‰ in preceding and following temperature steps, consistent with the pyrolysis technique enhancing resolution from isotopically light nitrogen in a host of approximately similar resistance to combustion. Figure 6 shows that there is no evidence for such isotopically heavy nitrogen in the other enstatite chondrites during stepped pyrolysis, and thus the effect would appear confined to Abee. In this respect, it must be noted that the material under investigation is not entirely representative of the meteorite, but is from a clast within the sample. The abruptness of the release, coupled with observation that it is effected by both combustion and pyrolysis at the same temperature, possibly implies a component which decrepitates, melts or degases over a very narrow temperature interval. Organic material seems unlikely, because the absence of oxygen gas does not elevate or broaden the release temperature; however it must be pointed out that there is a sharp release of carbon at 500–550°C during combustion of Abee.

Isotopically heavy nitrogen has been isolated in other groups of meteorites: (i) carbonaceous chondrites Al Rais, Bells, Kaidun, Renazzo, Yamato 790003 and Yamato 790112, which show δ^{15}N from +140 to +335‰ (KUNG and CLAYTON, 1978; ROBERT and EPSTEIN, 1982; GRADY *et al.*, 1983; KERRIDGE, 1985). The temperatures at which isotopically heavy nitrogen

is released from these meteorites are generally higher and more extended, therefore it is not clear what relationship (if any) exists to the nitrogen in Abee. (ii) in glass from EETA 79001 shergottite (BECKER and PEPIN, 1984) where it is believed to be relict trapped martian atmosphere. Nitrogen in SNC meteorites is confined to high temperature fractions, and is unlikely to be related to that found in Abee. (iii) in the unusual polymict breccias Bencubbin and Weatherford, which have bulk $\delta^{15}N$ values in excess of +900‰ (PROMBO and CLAYTON, 1985a,b; FRANCHI *et al.*, 1985). Although the major nitrogen-bearing components in the Bencubbin breccia liberate ^{15}N-enriched nitrogen above 900°C, a chondritic inclusion and an HCl-resistant residue from a metal clast release nitrogen with $\delta^{15}N$ up to +984‰ at temperatures as low as 400°C during combustion or 500°C on pyrolysis (FRANCHI *et al.*, 1985). Furthermore, pyrolysis of the chondritic xenolith provides nitrogen with $\delta^{15}N$ of +450‰ at 300–400°C (I. A. FRANCHI, unpublished data). Petrological comparisons between the Abee and Bencubbin clasts would be desirable, and an investigation of acid-insoluble residues from the Abee clast might be informative.

It is interesting to speculate on the significance of the isotopically heavy nitrogen in Abee. If it is indeed associated with the carbonaceous material which combusts over a similar, narrow temperature range, then it is possible that this is an unusual component, possibly "mysterite," which represents the residuum after condensation of other, more refractory components. It is yet another instance of isotopically "normal" carbon associated with extreme nitrogen isotopic compositions (LEWIS *et al.*, 1983). Since there is evidence that the Abee parent has been thermally metamorphosed to temperatures > 840°C (RUBIN and KEIL, 1983), a carrier of nitrogen with unusual $\delta^{15}N$, stable only to 500°C, must have been incorporated into the meteorite at a later stage in the meteorite's history, possibly during the brecciation and subsequent reconsolidation events.

Nitrogen in the Bustee aubrite (Fig. 4) is less abundant than in enstatite chondrites, and is dominantly locked up in osbornite ($\delta^{15}N$ = −24‰). There is little evidence of the presence of sinoite. Nitrogen released by combustion between 200–600°C from the residue of the sample hand-picked for osbornite has variable $\delta^{15}N$, and probably derives from a mixture of contaminants.

SUMMARY

There are several components of indigenous carbon in enstatite chondrites. The most abundant is that which combusts at temperatures above 600°C, primarily from elemental carbon as graphite, present in silicate grains and with metal. The concentration and isotopic composition of high temperature carbon is variable, but generally higher in type EL6 (0.16 to 0.51 wt%; $\delta^{13}C$ = −6.7 to +4.2‰) than EH4,5 (0.03 to 0.24 wt%; $\delta^{13}C$ = −9.1 to −5.8‰). Part of this variation may be due to the existence of small amounts of cohenite, also known to occur in enstatite chondrites, and which combusts at high temperatures. However, the exact abundance and isotopic composition of cohenite in these meteorites is unknown, although it seems likely that it is isotopically lighter than graphite.

Also combusting at elevated temperatures is another minor carbonaceous component with a highly unusual $\delta^{13}C$ value. This so-called "interstellar carbon" has been identified in acid-resistant residues from one EH3 and one EH4 chondrite, and will be present in low concentrations in at least some of the whole-rock samples. The high positive $\delta^{13}C$ signature of the material is sufficient to cause differences in carbon isotopic composition at the highest temperatures, even when the component occurs in very low abundances.

In addition to the above-mentioned components which combust at temperatures > 600°C, there is another major carbonaceous material which burns at lower temperatures. This also is present in variable abundance in both EH4,5 and EL6 meteorites, and possesses an isotopic composition *ca* −20 to −15‰, similar to that of the macromolecular material in carbonaceous chondrites, although at least a proportion of it may be of terrestrial derivation. Underlying this component is a second minor phase with $\delta^{13}C$ values < −40‰, the putative carrier of Xe (HL).

Nitrogen is present in a variety of forms; petrographic observation suggests that most probably occurs in the refractory minerals sinoite and osbornite, which both combust across a similar temperature range above 800°C. Since sinoite and osbornite combust so closely, it is difficult to place limits on their apparent abundances and $\delta^{15}N$ values, although it seems that sinoite is isotopically lighter than osbornite.

Organic nitrogen is also present in enstatite chondrites, again with variable abundance and isotopic composition, not apparently related to petrologic type. Part of the cause of the variability is, as for carbon, the presence of minor nitrogen-bearing phases. The Xe (HL) host has an isotopic signature depleted in ^{15}N, and in addition to this, in Abee there is an ^{15}N-enriched component, possibly associated with isotopically normal carbon, which breaks down at temperatures *ca.* 500°C. This latter component might be connected with "mysterite," the late-stage, volatile-rich component postulated to be present in Abee (GANAPATHY and LARIMER, 1980).

Carbon in the Bustee aubrite is less abundant and isotopically lighter than carbon in enstatite chondrites. It occurs as graphite (57 ppm, $\delta^{13}C$ *ca.* −22‰), although cohenite is possibly present (17 ppm with $\delta^{13}C$ = −24‰). However, most of the carbon in Bustee is material which combusts at low temperatures (160 ppm at 500°C, $\delta^{13}C$ *ca.* −24‰), tentatively identified as terrestrial organic contamination. There is no evidence of any highly ^{13}C-enriched or ^{13}C-depleted components. Nitrogen in Bustee occurs mostly as osbornite ($\delta^{15}N$ *ca.* −24‰); sinoite appears absent. Nitrogen which is not bound up in TiN is present as low temperature

organic material. Consistent with carbon measurements, there is no evidence for isotopically light or heavy nitrogen, indicating that the unusual minor components identified in enstatite chondrites are absent from Bustee.

The carbon and nitrogen data reported herein are certainly intriguing in the implications they may have for enstatite meteorite formation. The dissimilar values of $\delta^{13}C$ for the same phase (*viz* graphite) and its variable abundance in the three groups (EH4,5, EL6 and aubrites) points towards different origins for these meteorites, as the fractionation in carbon values cannot easily be explained in terms of progressive thermal metamorphism. Nitrogen abundance and isotope data are variable due to heterogeneity in distribution of discrete nitrogen-bearing minerals.

Acknowledgements—Gratitude is extended to the following who generously provided samples: Dr S. O. Agrell, Dr R. Hutchison, Dr K. MacNamara and Dr K. Marti. Drs R. H. Becker and D. W. Sears provided thorough, helpful and thought-provoking reviews which substantially clarified the manuscript, for which the authors are very grateful. We thank Mr I. A. Franchi for the use of unpublished data. Financial support was made available by the Science and Engineering Research Council.

Editorial handling: J. R. O'Neil

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